

Environmental Chemistry

1. Introduction

Environmental chemistry is the study of chemical processes occurring in the atmosphere, hydrosphere, lithosphere, and biosphere, and of the chemical effects of human activities on these systems. It is an interdisciplinary subject linking chemistry, ecology, toxicology, geology, and public health.

For B.Sc. Chemistry students, it is important because it explains pollution, monitoring methods, water and air quality, waste treatment, climate change, and environmental protection laws.

2. Scope and importance

Environmental chemistry helps us understand the sources, reactions, transport, effects, and control of pollutants. It also provides the chemical basis for clean water, clean air, soil conservation, waste management, and sustainable development.

The subject has major applications in environmental monitoring, industry, agriculture, public health, and policy.

3. Environmental segments

The environment is commonly divided into four major segments:

- Atmosphere.
- Hydrosphere.
- Lithosphere.
- Biosphere.

Atmosphere

The atmosphere is the gaseous envelope around Earth and is essential for life, weather, and climate regulation.

Hydrosphere

The hydrosphere includes oceans, rivers, lakes, groundwater, and other water bodies.

Lithosphere

The lithosphere includes soil, rocks, minerals, and the Earth's solid outer layer.

Biosphere

The biosphere includes all living organisms and their interactions with the environment.

4. Biogeochemical cycles

Biogeochemical cycles describe the circulation of elements between living and non-living components of the Earth.

Important cycles

- Carbon cycle.
- Nitrogen cycle.
- Sulfur cycle.
- Phosphorus cycle.
- Oxygen cycle.

Importance

These cycles maintain ecological balance and determine the distribution and availability of essential elements. Human activities can disturb them and cause pollution or climate effects.

5. Air pollution

Air pollution is the presence of harmful substances in the atmosphere at concentrations that adversely affect health, vegetation, animals, materials, and climate.

Major air pollutants

- Particulate matter.
- Carbon monoxide.
- Carbon dioxide.
- Sulfur dioxide.
- Nitrogen oxides.
- Hydrogen sulfide.
- Hydrocarbons.
- Lead compounds.
- Photochemical oxidants.

Sources

- Vehicular emissions.
- Thermal power plants.
- Industrial combustion.
- Refineries.

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- Biomass burning.
- Construction dust.
- Domestic fuel use.

Effects

- Respiratory diseases.
- Eye and throat irritation.
- Reduced plant productivity.
- Corrosion of materials.
- Global warming.
- Ozone depletion.
- Acid rain.

6. Atmospheric chemistry

The atmosphere undergoes numerous chemical and photochemical reactions involving oxygen, nitrogen oxides, sulfur compounds, and hydrocarbons. These reactions are responsible for smog, ozone formation and depletion, and secondary pollutant production.

Important processes

- Photolysis.
- Radical chain reactions.
- Oxidation by hydroxyl radicals.
- Smog formation.
- Ozone chemistry.

7. Photochemical smog

Photochemical smog forms when sunlight acts on nitrogen oxides and hydrocarbons, producing ozone and other oxidants. It is common in urban areas with heavy traffic and industrial emissions.

Constituents

- Ozone.
- Peroxyacyl nitrates.
- Aldehydes.
- Nitrogen dioxide.
- Secondary organic aerosols.

8. Acid rain

Acid rain is precipitation with abnormal acidity caused mainly by sulfur dioxide and nitrogen oxides forming sulfuric and nitric acids in the atmosphere.

Effects

- Damage to lakes and aquatic life.
- Soil acidification.
- Corrosion of buildings and monuments.
- Damage to crops and forests.

9. Ozone layer depletion

The ozone layer protects life from harmful ultraviolet radiation. Ozone depletion is mainly caused by chlorine- and bromine-containing compounds such as CFCs and related substances.

Effects

- Skin cancer risk.
- Cataracts.
- Damage to phytoplankton.
- Reduced crop yields.
- Ecosystem imbalance.

10. Greenhouse effect and global warming

The greenhouse effect is the trapping of outgoing infrared radiation by greenhouse gases such as carbon dioxide, methane, nitrous oxide, and water vapor. Global warming is the long-term rise in average Earth temperature due to enhanced greenhouse effect.

Effects

- Climate change.
- Sea-level rise.
- Extreme weather events.
- Melting glaciers.
- Ecosystem shifts.

11. Water pollution

Water pollution is the contamination of water bodies by physical, chemical, or biological pollutants that degrade water quality and harm living organisms.

Sources

- Domestic sewage.
- Industrial effluents.
- Agricultural runoff.
- Oil spills.
- Thermal discharges.
- Heavy metals.
- Detergents and pesticides.

Effects

- Eutrophication.
- Oxygen depletion.
- Spread of waterborne diseases.
- Bioaccumulation of toxins.
- Damage to aquatic ecosystems.

12. Water quality parameters

Water quality is assessed using physical and chemical parameters.

Important parameters

- pH.
- Turbidity.
- Dissolved oxygen.
- BOD.
- COD.
- Suspended solids.
- Hardness.
- Alkalinity.
- Chlorides.

13. Water treatment

Water treatment converts contaminated water into safe water for drinking or industrial use.

Methods

- Sedimentation.
- Filtration.
- Chlorination.
- Reverse osmosis.
- Ion exchange.
- Electro-dialysis.
- Softening.
- Aeration.

14. Hardness of water

Hardness is caused mainly by calcium and magnesium salts. It may be temporary or permanent.

Removal methods

- Boiling.
- Lime-soda process.
- Zeolite process.
- Ion exchange method.

15. Soil pollution

Soil pollution is the degradation of soil quality by chemical contaminants, salts, heavy metals, pesticides, waste dumping, and industrial residues.

Causes

- Excess fertilizers.
- Pesticides.
- Industrial waste.
- Mining.
- Plastic waste.
- Sewage sludge.
- Oil contamination.

Effects

- Loss of fertility.
- Toxicity to plants.
- Reduced crop quality.
- Groundwater contamination.
- Harm to soil biota.

16. Soil and water analysis

Environmental chemistry often includes analysis of soil and water samples to determine contamination and nutrient status.

Common tests

- pH.
- Conductivity.
- Hardness.
- Chloride.
- Turbidity.
- Ca and Mg content.
- Organic carbon.
- Nutrients.

17. Toxic chemicals

Toxic chemicals in the environment include heavy metals, cyanides, pesticides, solvents, and industrial chemicals.

Heavy metals

- Lead.
- Mercury.
- Cadmium.
- Arsenic.

Effects

- Nervous system damage.
- Kidney damage.
- Bone disorders.

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- Cancer risk.
- Developmental toxicity.

18. Pesticides

Pesticides are chemicals used to control pests, but they may cause environmental and health problems through persistence, bioaccumulation, and food chain transfer.

19. Solid waste management

Solid waste management includes collection, segregation, recycling, treatment, and safe disposal of municipal and industrial waste.

Methods

- Recycling.
- Composting.
- Incineration.
- Sanitary landfilling.
- Segregation at source.

20. Industrial waste and effluents

Industrial effluents from textile, tannery, electroplating, dairy, fertilizer, petroleum, and agro-industries require treatment before release.

Treatment stages

- Primary treatment.
- Secondary biological treatment.
- Tertiary or advanced treatment.

21. Nuclear pollution

Nuclear pollution arises from radioactive waste, nuclear accidents, weapons testing, and improper disposal of radioactive materials.

Problems

- Radiation exposure.
- Genetic damage.
- Long-term contamination.
- Disposal difficulty.

22. Energy and environment

Energy use is closely tied to environmental quality. Conventional sources such as coal, petrol, and natural gas contribute significantly to pollution, while alternative sources reduce environmental load.

Sources

- Solar energy.
- Wind energy.
- Hydroelectric energy.
- Biomass energy.
- Nuclear energy.
- Hydrogen energy.
- Fusion research.

23. Environmental laws

Environmental chemistry also covers laws and regulations that protect air, water, soil, wildlife, forests, and public health.

Examples

- Environment Protection Act.
- Air Prevention and Control of Pollution Act.
- Water Prevention and Control of Pollution Act.
- Wildlife Protection Act.
- Forest Conservation Act.
- Ozone-related regulation.

24. Environmental disasters

Major disasters are often studied because they show the consequences of chemical pollution and poor industrial safety.

Examples

- Bhopal gas disaster.
- Seveso disaster.
- Chernobyl incident.
- Minamata mercury poisoning.
- Cadmium poisoning in Itai-itai disease.

25. Comparisons

Air pollution vs water pollution

- Air pollution affects atmosphere and respiration.
- Water pollution affects aquatic systems and drinking water.
- Both can originate from industry, transport, and human activity.

BOD vs COD

- BOD measures biodegradable oxygen demand.
- COD measures total chemically oxidizable matter.
- COD is usually higher than BOD.

Temporary hardness vs permanent hardness

- Temporary hardness is removed by boiling.
- Permanent hardness needs chemical treatment.

26. Summary

Environmental chemistry explains how chemicals move through air, water, soil, and living systems, and how human activities alter those systems. For B.Sc. Chemistry students, it is a highly applied subject that links chemistry to pollution control, public health, waste treatment, and sustainable environmental management.